Quantification: 535_z13c1-2
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	10
interior lacuna count	10
frame-edge lacuna count	0

Interior lacunae only: mean, median and sample SD (n = 10)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	3598.7	3288	2943.18	px^2
major_axis_length_px	lacuna shape	98.5303	100.829	49.7684	px
minor_axis_length_px	lacuna shape	47.4698	44.6715	22.3445	px
aspect_ratio	lacuna shape	2.0393	2.0322	0.3787	unitless
eccentricity	lacuna shape	0.8511	0.8703	0.0825	unitless
solidity	lacuna shape	0.7983	0.7918	0.0958	unitless
roots_count	headline	9.4	10	5.379	count
ring_length_r30_px	headline	310.3	262	154.307	px
ring_length_r60_px	reported	628.9	521	291.231	px
owned_length_px	ownership-dependent	733.646	508.268	613.493	px
edge_count	ownership-dependent	35.2	25	30.6841	count
mean_edge_length_px	ownership-dependent	20.21	20.3485	5.9939	px
perimeter_px	normalised	321.941	291.392	226.748	px
ring_area_r30_px2	normalised	10849	10614	4854.21	px^2
ring_area_r60_px2	normalised	23894	23699.5	8877.75	px^2
in_frame_fraction_r30	normalised	1	1	0	unitless
in_frame_fraction_r60	normalised	0.9989	1	0.0036	unitless
ring_density_r30	normalised	0.0284211	0.0269633	0.00760366	px^-1
ring_density_r60	normalised	0.0262811	0.0241576	0.00742116	px^-1
roots_per_100px_perimeter	normalised	3.2451	3.2284	1.9545	per 100 px
ring_attached_length_r30_px	network v2	274.2	236.5	159.861	px
ring_attached_length_r60_px	network v2	483.4	374	330.06	px
ring_length_w_r30_px	network v2	349.7	295.693	178.032	px
ring_length_w_r60_px	network v2	710.455	585.452	335.7	px
sholl_crossings_r10	network v2	7.8	7	3.6757	count
sholl_crossings_r20	network v2	8.1	8	3.5418	count
sholl_crossings_r30	network v2	8.8	7.5	3.9101	count
loop_count	loops, report only	1.6	1	2.1705	count
loop_edge_share	loops, report only	0.0958	0.1	0.0935	share
ring_width_mean_r30_px	width	7.3514	7.5896	1.208	px

Per field
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measure	value	units
analysed_area_px2	1.01259e+06	px^2
total_skeleton_length_px	21330	px
canalicular_length_density_per_px	0.0210648	px^-1
junction_count	369	count
junction_density_per_px2	0.00036441	px^-2
lacuna_count	10	count

lacunae_per_px2	9.88e-06	px^-2
skeleton_component_count	349	count
mean_component_length_px	61.1175	px
median_component_length_px	19	px
field_length_density_w_per_px	0.0236803	px^-1
field_density_without_flagged_per_px	0.0205768	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	782	px
unattached_component_count	48	count
canalicular_length_density_attached_per_px	0.0210648	px^-1
field_length_density_with_unattached_per_px	0.0218371	px^-1
excluded_area_px2	0	px^2
analysed_field_area_px2	1.04858e+06	px^2
loop_count	28	count
loop_edge_share	0.0863	share
edge_count	1078	count

Canalicular width, the new measure

measure	value	units
width_mean_px	7.1946	px
width_median_px	7.2111	px
width_p10_px	4	px
width_p90_px	10.7703	px
width_pixels_used	19774	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel.
Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the
dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the
pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every
thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	3449	94.2445	51.4301	1.8325	0.838	0.8008	-1.3442	87.5868	746.098	False
2	916	42.5998	32.8395	1.2972	0.637	0.7829	0.4554	434.044	377.415	False
3	6729	150.978	65.1171	2.3186	0.9022	0.7721	-0.3046	496.001	291.95	False
4	3320	105.162	44.1209	2.3835	0.9077	0.7721	-1.4464	464.635	577.918	False
5	3645	102.737	49.6538	2.0691	0.8754	0.8697	0.3597	580.157	513.438	False
6	2801	98.9222	38.3287	2.5809	0.9219	0.8173	0.524	580.027	151.776	False
7	10263	199.04	99.7612	1.9952	0.8653	0.5873	-0.102	659.792	674.277	False
8	3256	105.273	45.222	2.3279	0.903	0.7485	-1.2191	649.232	325.526	False
9	998	48.4903	27.4457	1.7668	0.8244	0.8951	1.3306	782.386	307.599	False
10	610	37.8566	20.7792	1.8218	0.8359	0.937	-0.7458	884.313	871.107	False

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	9	393	1014	1592.07	82	19.4154	7.1937
2	11	234	429	769.997	33	23.3332	7.4626
3	17	485	939	1529.41	84	18.2072	8.2352
4	7	261	416	225.024	12	18.752	6.6633
5	3	204	417	202.64	7	28.9485	8.5039
6	12	263	534	499.031	24	20.793	7.9127
7	16	601	1147	1591.01	65	24.4771	7.5101

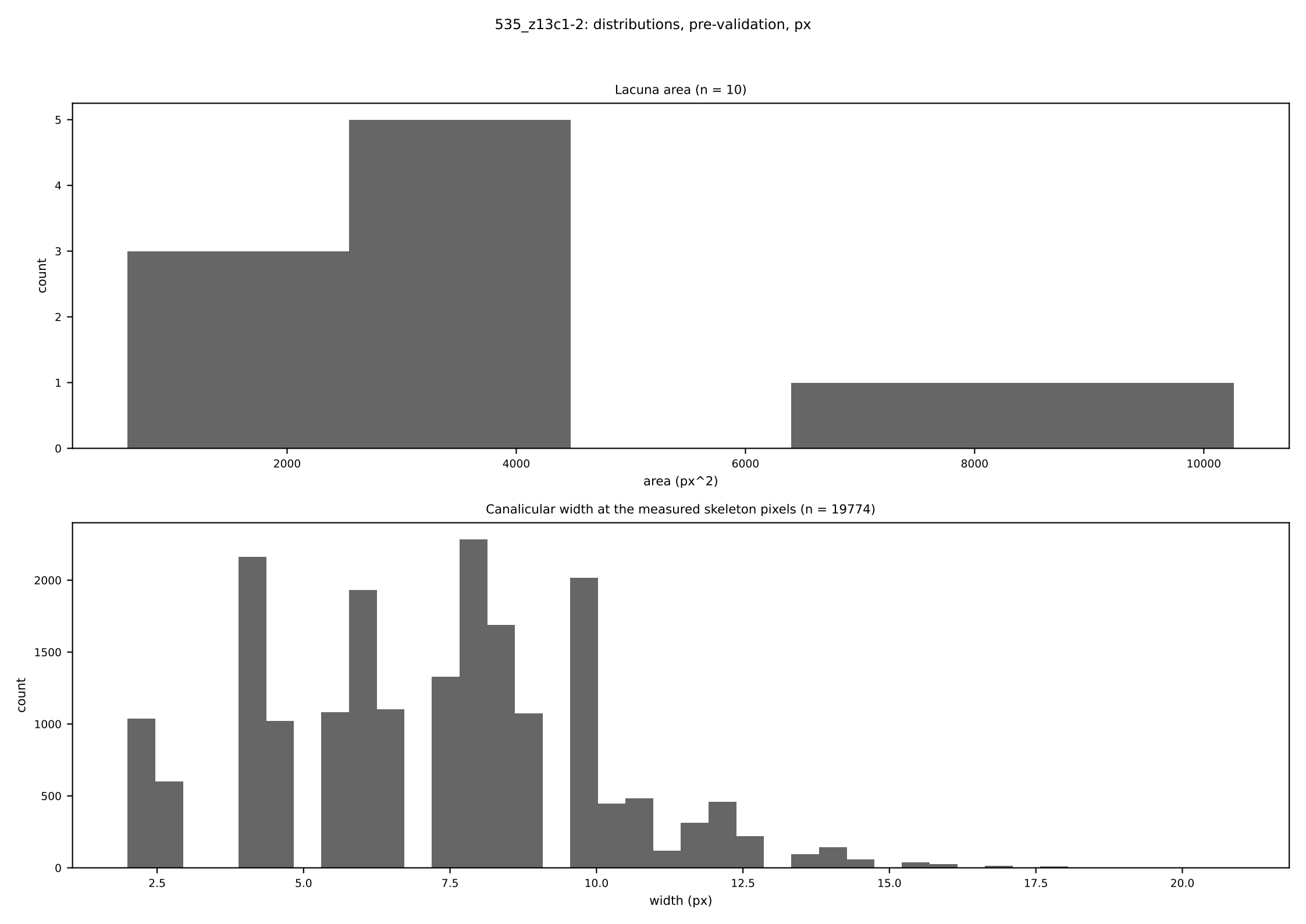
8	13	390	559	403.948	18	22.4416	7.669
9	5	178	508	517.505	26	19.904	8.0968
10	1	94	326	5.8284	1	5.8284	4.2665

Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	311.848	10999	27029	1	0.9885	0.0357305	0.0375153	2.886
2	145.782	5773	13561	1	1	0.0405335	0.0316348	7.5455
3	433.103	13694	24741	1	1	0.035417	0.0379532	3.9252
4	326.563	11164	23645	1	1	0.0233787	0.0175936	2.1435
5	270.936	10229	21506	1	1	0.0199433	0.0193899	1.1073
6	264.007	10205	23754	1	1	0.0257717	0.0224804	4.5453
7	897.46	22322	46257	1	1	0.0269241	0.0247962	1.7828
8	328.877	11820	23768	1	1	0.0329949	0.023519	3.9528
9	140.024	6592	17645	1	1	0.0270024	0.02879	3.5708
10	100.811	5692	17034	1	1	0.0165144	0.0191382	0.992

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	385	868	442.09	1162.02	10	9	12
2	224	387	262.066	490.517	7	6	6
3	452	894	551.931	1065.34	13	10	14
4	189	210	288.309	456.777	6	12	6
5	186	318	226.095	460.948	7	8	5
6	249	430	303.078	603.115	7	8	8
7	572	1030	682.642	1302.37	14	14	15
8	316	361	447.475	633.428	8	8	11
9	137	304	191.711	567.789	3	4	7
10	32	32	101.598	362.25	3	2	4


535_z13c1-2: distributions, pre-validation, px
Lacuna area (n = 10)
count
area (px^2)
0
1
2
3
4
5
2000
4000
6000
8000
10000
Canalicular width at the measured skeleton pixels (n = 19774)
count
width (px)
0
500
1000
1500
2000
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0